Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	04 April 2026
Team ID	SWTID-2026-4754
Project Name	Automating Employee Data
Maximum Marks	4 Marks

1. Technical Architecture

The system follows a 3-tier architecture consisting of:

- Presentation Layer (UI) – Handles user interaction
- Application Layer (Backend) – Processes business logic
- Data Layer (Database) – Stores and retrieves data

Architecture Description:

- Users interact through a web or mobile interface
- Requests are sent to the backend server
- Backend processes logic and communicates with database
- External APIs may be used for authentication and services
- Data is stored securely in cloud/local databases

2. Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for user interaction	HTML, CSS, JavaScript, React JS
2	Application Logic-1	Handles user registration & login	Java / Python (Django / Spring Boot)
3	Application Logic-2	Handles email/OTP verification	Node.js / Firebase Authentication
4	Application Logic-3	Business logic for employee management	Python / Java
5	Database	Stores employee and user data	MySQL / MongoDB
6	Cloud Database	Cloud-based storage	Firebase / AWS RDS
7	File Storage	Stores documents and files	AWS S3 / Local Storage
8	External API-1	Email & OTP services	Gmail API / Twilio API
9	External API-2	Authentication services	Google OAuth / LinkedIn API
10	Machine Learning Model	(Optional) Data analysis or predictions	Python ML Model
11	Infrastructure	Deployment environment	AWS / Azure / Local Server

---

3. Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for development	React JS, Django, Node.js
2	Security Implementations	Data protection and access control	HTTPS, JWT, SHA-256, OAuth
3	Scalable Architecture	System can handle increasing users	Microservices / 3-tier architecture
4	Availability	System accessible anytime	Cloud hosting, Load Balancer
5	Performance	Fast response and efficient system	Caching (Redis), CDN

---

4. Conclusion

The chosen technology stack ensures:

- Efficient system performance
- High security and reliability
- Scalability for future growth
- Easy maintenance and deployment

This architecture provides a strong foundation for building a robust and user-friendly employee data management system.